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Recognition Performance Validation of Weather Map Images by ChatGPT

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Abstract

Weather map recognition is a complex image recognition task that requires two cognitive processes: the interpretation of symbols and future predictions. Recent advancements in multimodal AI have shown the potential to solve such complex tasks. This research uses ChatGPT, a type of multimodal AI, to validate the recognition performance of weather map images and proposes prompt engineering for more accurate recognition. Weather map recognition is a problem that is also featured in the Common University Entrance Test. If weather maps can be recognized, automatic grading and question generation become possible, leading to learning support. Furthermore, it is expected that weather commentary text can be automatically generated from weather maps. Therefore, weather map image recognition is an important task, but it requires not only recognizing time-series changes in local weather data but also recognizing information on large-scale weather patterns such as pressure pattern and making future predictions. In this research, we use ChatGPT4-Vision, a multimodal AI, to validate the recognition performance of weather map images. We investigate whether it can answer questions about weather maps from university entrance exams and generate weather commentary text from weather map images. By conducting multiple experiments with varying tasks and information in the prompts, and evaluating the accuracy of the generated commentaries, we propose and validate improvements in reading performance through prompt engineering.

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1. Introduction

The purpose of this research is to investigate the recognition performance of weather map images using ChatGPT, a multimodal AI, and to propose prompt engineering methods for more accurate recognition, thereby contributing

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to the development of AI-assisted weather map interpretation and commentary generation to support education and personalized weather forecasts.

Understanding weather maps involves two cognitive processes: the basic process of accurately interpreting abstract shapes and symbols shown on the map, and the reasoning process of making future predictions based on this information [3]. Therefore, not only does this task require the interpretation of information summarized in diagrams, but it also necessitates making future forecasts, making it a more challenging task than general image recognition.

Recently, with the advancement of multimodal AI, the potential to solve such complex image recognition tasks has been demonstrated [5]. In this research, we use one type of multimodal AI, ChatGPT, to verify the recognition performance of weather map images and propose prompt engineering for more accurate recognition.

Weather map recognition is tested in the "Common University Entrance Test," a standardized exam used by universities in Japan. Being able to recognize weather maps could facilitate automatic grading and question generation, thereby supporting student learning. Furthermore, it is believed that recognizing weather maps can enable the automatic generation of weather commentary texts. These commentaries are produced by meteorologists who interpret and verbalize information from weather maps and satellite images. This task requires creating different texts based on the medium and the region addressed, and there is a limit to the number of commentaries that can be produced within a set time. If weather commentaries could be generated automatically without human intervention, it would be possible to deliver customized weather forecasts to every citizen.

Historically, research on the automatic generation of weather commentaries has focused on creating structured data such as tables and databases from numerical weather prediction data, based on expert meteorological knowledge, and then generating text from this data [2, 1, 12]. Recently, with improvements in computational power allowing for increased data input, studies have also been conducted that directly input numerical weather prediction data to generate text [8]. There are two main challenges in this research. The first challenge is to include information about large-scale weather patterns, such as pressure systems and high/low pressure areas, in the commentaries, not just time-series changes in local weather data. The second challenge is to generate text that is natural and free of awkward phrasing.

To address the first challenge, large-scale weather patterns need to be interpreted from surface weather map images, a task that is also a common issue in the university entrance exams. This line of research has traditionally been approached by calculating similarity based on pressure pattern [9]. More recent studies have moved beyond simple pixel array comparisons to calculate similarities by segmenting the images into layers representing different meteorological elements such as isobars and pressure systems [14]. This segmentation allows for the interpretation of the information contained in the images as a type of object recognition task, which has seen significant performance improvements with the advent of the Transformer model, which efficiently captures relationships between data using the Attention mechanism [15].

The second challenge, generating natural text, has seen breakthroughs due to large language models (LLMs) like ChatGPT [17]. Additionally, in 2023, multimodal AIs capable of generating text from images were released by organizations like OpenAI and Google [7]. Specifically, OpenAI's ChatGPT4-Vision [10] has shown high performance in a variety of tasks including image generation from text, table data analysis, speech recognition, and image interpretation. Thus, multimodal AIs hold the potential to simultaneously address both the interpretation of meteorological fields and the generation of natural text.

To utilize ChatGPT for specific tasks, it is essential to input a "prompt," a command that informs the AI of the expected output characteristics and format. Therefore, the prompt is crucial for optimizing performance and obtaining more relevant outputs. Since the release of ChatGPT, various insights into prompt engineering have been discovered. For example, structuring prompts and clarifying instructions can enhance performance [16]. Additionally, when tasks involve multiple procedures, providing a 'Step-by-step' instructional prompt ensures the instructions can be executed reliably [4]. Furthermore, incorporating a few example sentences in the prompt, a method known as "Few-shot learning," helps ChatGPT understand new concepts and patterns, facilitating generalization [13].

This research employs ChatGPT4-Vision, a multimodal AI, to verify the recognition performance of surface weather map images. We test whether it can answer questions from the Common University Test about surface weather maps and generate weather commentary from these images. By varying the tasks and the amount of information provided in the prompts across different experiments, and by evaluating the accuracy of the generated commentaries, we propose and verify improvements in reading performance through prompt engineering.

2. Experiments to Validate University Entrance Exam

2.1. Conditions and methods

We extracted five questions on reading surface weather maps from the Earth Science Basics section administered in the Common Test for University Admissions¹, specifically from the academic years 2019, 2022, 2023, and 2024, to evaluate the reading capabilities of ChatGPT-4 Vision.

We used the "gpt-4-1106-preview" model of the ChatGPT4-Vision API to input the images and prompts that included the question text. The responses were obtained in the form of numbers 1 to 4, from which a correct or incorrect answer was determined.

The images input were of weather-related questions from the Japanese Common Test for University Admissions. Since these are Common Test questions, the output of answers is always in the form of numbers 1 to 4. Please follow the step-by-step process below to answer the question and output the answer.

#Process

step0. Convert the image into text.

step1. Read the surface weather map and text within the image to understand the question.

step2. Read the surface weather map and text within the image to understand how to respond.

step3. Considering the question and how to respond, output the answer.

step4. Explain in Japanese why you think that answer is correct.

2.2. Results

The experimental results for the five questions are shown in the table 1 .

Table 1. Experiments to Validate University Entrance Exam Results

Year	2019	2022	2022	2023	2024
Question Part	Part 2	Part 2	Part 2	Part 2	Part 2
Question Number	Q1	Q1	Q2	Q1	Q1
Experiments to Validate University Entrance Exam	Correct	Correct	Incorrect	Correct	Correct

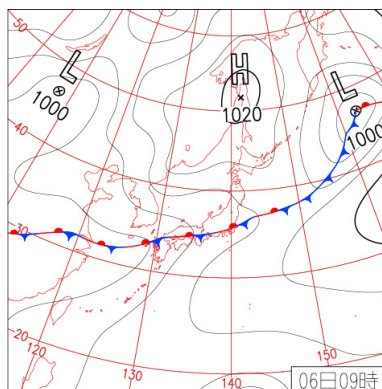


Fig. 1. Surface Weather map at 9 AM on May 1, 2022.

¹ The university entrance qualification examination in Japan, where the test questions are provided by the National Center for University Entrance Examinations.

For the targeted questions, ChatGPT-4 Vision was able to correctly answer four out of five questions. For instance, it correctly identified the wind direction on both sides of the seasonal rain front in the second question of 2022, Fig. 1. From the questions it answered correctly, ChatGPT-4 Vision demonstrated the ability to compare surface weather maps from different times, accurately capturing the movement and development of typhoons. It also succeeded in reading the sea level pressure at a specific point from the isobars and deducing related meteorological information. These results indicate that ChatGPT-4 Vision is capable of comprehensively understanding meteorological information represented on surface weather maps and making appropriate judgments in response to the questions.

On the other hand, it answered incorrectly for question 1 of the second part of 2022. The question asked whether the area is wet or dry with respect to the high pressure of 1020 hPa within Fig. 1. It is assumed that the moist Okhotsk Sea High at 1020 hPa within Fig. 1 was misidentified as a continental air mass, and thus misidentified as a dry Siberian air mass. This suggests that the distinction between isobars and coastlines could not be discerned from the surface weather maps image of the question. Therefore, it is considered necessary to devise ways to ensure recognition of the land which serves as a reference point for the location of high pressure. The task required predicting detailed meteorological conditions at a specific location based on information about pressure pattern and fronts. While ChatGPT-4 Vision was able to infer the general meteorological situation from the information provided, it struggled with predicting conditions for specific locations. These findings suggest limitations in the information that ChatGPT-4 Vision can extract from surface weather maps, indicating that more detailed information is needed for precise weather prediction.

3. Experiments to Validate weather Commentary Generation

3.1. Conditions and methods

For the input data, we utilized surface weather map images published on the Japan Meteorological Agency website in the "Daily Weather Maps" section (available at: <https://www.data.jma.go.jp/yoho/hibiten/index.html>, accessed on March 2, 2024). Additionally, in experiments where meteorological data were added to the prompts, we input 16 types of measurements for 56 locations including Regional and Local Meteorological Observatories. These measurements included daily precipitation, maximum one-hour precipitation, maximum ten-minute precipitation, average temperature, maximum temperature, minimum temperature, average humidity, minimum humidity, average wind speed, maximum wind speed, direction of maximum wind speed, maximum gust speed, direction of maximum gust speed, sunshine duration, snowfall, and maximum snow depth.

The difficulty of generating comments is thought to vary depending on weather conditions. Therefore, the target cases were selected as the 1st day of each month from January to December 2022, as well as June 21st when a rainy season front was present on the weather map, September 17 when a typhoon existed, and January 6 when snowfall occurred in the Kanto region due to the influence of a south coast low pressure system, for a total of 15 days. In the experiment where example sentences were added to the output, in order to adapt to diverse weather conditions, the weather overviews for the 1st day of each month in 2020 and a total of 15 days when a rainy season front, typhoon, or south coast low pressure system were present, as shown on the daily weather maps, were used as input.

We conducted four types of weather commentary generation experiments using the "gpt-4-1106-preview" model, an API of ChatGPT4-Vision. We input surface weather map images and prompts into the model. To clarify the instructions, titles for the directives were given using Markdown notation in all experiments. Among these, Experiment 1 employs the simplest method, specifying character count and expression style in the prompt as output conditions.

Please generate a weather commentary based on the input weather map under the following conditions:

- The commentary should employ many nominal phrases, echoing the tone of an expert.
- This text should represent the current meteorological conditions depicted in the weather map rather than forecasts; hence, avoid ambiguous expressions such as "might", "could be", or "appears to be". Statements should be definitive.
- Ensure that the commentary is at least 80 characters long, counting punctuations, hiragana, and kanji as one character each.

- The commentary should be in Japanese.
- The date does not need to be included in the commentary.
- Do not include any irrelevant text outside of the meteorological information.

In contrast, Experiments 2 to 4 aim to improve performance by dividing the tasks into step-by-step processes and adding example sentences as outputs under the few-shot learning framework. Experiments 3 and 4 also include meteorological data from 56 locations relevant to the day in question. The prompt for Experiment 4, which involves the most tasks, is outlined below. It instructs to process the information step-by-step at the beginning of the prompt and clearly specifies the number of steps to be given.

Please process the following steps based on the input weather map image and meteorological data, step-by-step. There are a total of 11 steps. Make sure to complete all the steps. During the checking steps, confirm whether the previous step was indeed correct. Output should be in Japanese.

Following these instructions, 11 specific procedures are described. Here is a summary of each step:

1. Identify the type and location of air pressure and fronts in the image.
2. Checking of Step 1.
3. Analysis of the pressure pattern based on the positional relationships of pressure and fronts.
4. Analysis of the impact of the pressure pattern on Japan's major regions and the expected future impacts.
5. Checking of Step 4.
6. Creation of regional weather summaries based on numerical meteorological data.
7. Creation of weather summaries for each region.
8. Checking of Step 7.
9. Extraction of important meteorological conditions from numerical data.
10. Checking of Step 9.
11. Specification of the conditions for the document to be generated and generation of the commentary.

After the description of these 11 procedures, the text includes example outputs and meteorological data from 56 locations on the relevant day.

In Experiment 3, the procedure consists of 7 steps, excluding the checking tasks of Steps 2, 5, 8, and 10 from Experiment 4. In Experiment 2, as meteorological data are not inputted, the steps involving analysis of meteorological data, specifically Steps 6 and 9, are further excluded, resulting in a total of 5 steps. Summarizing these four experiments, the details are as outlined in Table 2. By comparing these experiments, it is possible to verify to what extent incrementally increasing tasks and information contributes to the improvement of commentary generation performance.

Table 2. experimental conditions

	Step-by-step	Example Sentences	Meteorological Data	Checking Operation
Experiment 1	-	-	-	-
Experiment 2	✓	✓	-	-
Experiment 3	✓	✓	✓	-
Experiment 4	✓	✓	✓	✓

3.2. Evaluation metrics

In order to provide a multifaceted and accurate evaluation of the LLM output comments, this research employs two evaluation methods to evaluate the generated comments: automatic evaluation and human evaluation. In the automatic evaluation, the accuracy of the "weather map" was evaluated using BLEU [11], ROUGE-1 [6], and the cosine similarity of the embedded representation.

BLEU and ROUGE have long been used in the field of text generation as measures of word-level agreement and are considered effective metrics for document summarization tasks. However, they may not always be appropriate for evaluating text generation using LLMs. Therefore, an evaluation using cosine similarity was also performed, which considers the semantic similarity of the content described in the comments. This involved vectorizing the text using the embedding model and calculating the cosine similarity. The embedding model represents the meanings of sentences in vector space, and in this research, OpenAI's "text-embedding-3-large" model was used to produce 3,072-dimensional vectors for this purpose.

In the manual evaluation, two of the authors, both experienced meteorologists, compared the input surface weather map images with the generated comments to judge the accuracy of the information. Specifically, they identified incorrect information and parts that, although not incorrect, were deemed inappropriate for a weather commentary. These judgments were integrated with the judgments of all judges for the number of characters generated and evaluated quantitatively based on the following definitions.

$$\text{Accuracy Rate} = \frac{\text{Number of characters not judged as errors or inappropriate}}{\text{Total number of characters generated}} \quad (1)$$

$$\text{Misinformation Rate} = \frac{\text{Number of characters judged as errors}}{\text{Total number of characters generated}} \quad (2)$$

$$\text{Inappropriateness Rate} = \frac{\text{Number of characters judged as inappropriate but not as errors}}{\text{Total number of characters generated}} \quad (3)$$

Thus, the accuracy rate is defined as the proportion of characters that were not identified as incorrect or inappropriate by both evaluators. The misinformation rate is the proportion of characters that were deemed incorrect by either of the evaluators. The inappropriateness rate refers to the proportion of characters that were considered inappropriate by at least one evaluator, but were not identified as incorrect by anyone else.

3.3. Results

3.3.1. Statistical evaluation

The average evaluation scores obtained from the 15-day data are shown in Table 3. All these metrics are such that higher values indicate better performance.

Table 3. Average Automated Evaluation Scores

	BLEU	ROUGE-1	Embedding Cosine Similarity (OpenAI)
Experiment 1	2.95	0.24	0.56
Experiment 2	3.25	0.31	0.61
Experiment 3	3.16	0.29	0.63
Experiment 4	3.10	0.30	0.64

From Table 3, it is evident that Experiments 2 to 4, which involved more complex prompts, were able to generate higher quality comments compared to Experiment 1, which used the simplest prompt. Moreover, in Experiments 3 and 4, the BLEU and ROUGE-1 scores decreased compared to Experiment 2. This decrease, as also noted by Murakami et al. (2021), is thought to be due to the introduction of specific numerical information in the prompts, which increased the amount of information generated and consequently reduced the word agreement. There was little difference between Experiments 3 and 4. The presence or absence of a checking step to interpret weather maps by ChatGPT accounted for the difference between these experiments. To analyze this detail further, the results of the accuracy rate, misinformation rate, and inappropriateness rate in the human evaluation are shown in Table 4.

From Table 4, it is evident that the misinformation rate has decreased from Experiment 3 to 4. This suggests that the checking tasks have contributed, albeit slightly, to the correction of the interpreted information.

Table 4. Average Expert Evaluation Scores

	Accuracy Rate	Misinformation Rate	Inappropriateness Rate
Experiment 1	0.29	0.46	0.25
Experiment 2	0.37	0.52	0.10
Experiment 3	0.51	0.34	0.15
Experiment 4	0.49	0.32	0.19

3.3.2. Correction of errors through checking operation

To examine how errors are corrected through the checking tasks performed in Experiment 4, we analyze the case of May 1, 2022, where errors were indeed corrected following the checking task step. The Surface Weather map image for this day is shown in Figure 2.

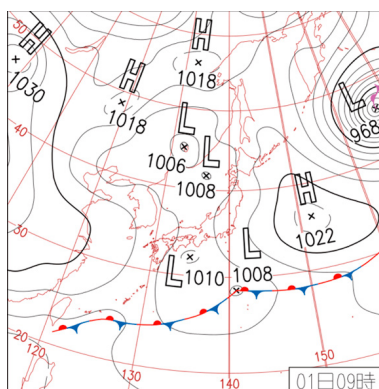


Fig. 2. Surface Weather map at 9 AM on May 1, 2022.

First, the output from Step 1, which involves reading the the central pressure of low pressures and front from Figure 2, is excerpted below:

”Over Japan: Low pressures of 1006hPa and 1008hPa are present.”

This accurately captures the central pressure of the low pressures near Japan. However, in Step 3, which analyzes the pressure pattern, an incorrect interpretation is made:

”There appears to be no direct impact of low pressure over Japan, but high pressure exists to the north and east, placing Hokkaido just south of the high pressure.”

This results in a loss of information from Step 1. This error is carried over into Step 4, which analyzes the effects of the pressure pattern. In contrast, Step 5, which checks errors using the numerical meteorological data inputted, corrects the inherited mistake:

”In the Kanto region, precipitation is generally as expected, but the decision that there were clear skies based on the pressure configuration was incorrect.”

This correction, made by assessing the presence of precipitation from the meteorological data, is evident in the final generated commentary:

“Japan has low pressure to the south and high pressure to the north. Hokkaido experiences strong winds with some clear spells. Kanto and Central Japan are cloudy with rain and high humidity. Generally good weather in Kinki and westwards, though locally unstable. Unstable weather continues in Okinawa.”

The corrected information regarding cloudiness and rain in Kanto and Central Japan is reflected in the generated commentary. This case demonstrates how inserting meteorological data into the prompt and implementing a checking step can correct the interpretation of the diagram and contribute to a reduction in the rate of misinformation.

However, there were instances where corrections suggested during the checking tasks were not reflected in the final generated commentary. An analysis of such a case on June 1, 2022, follows. The surface weather map image for this day is shown in Figure 3.

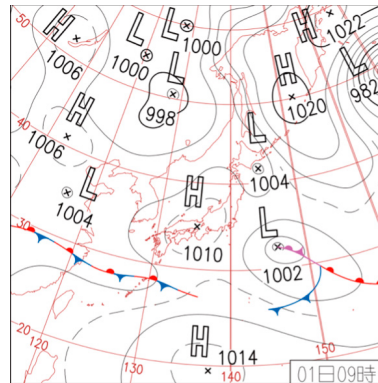


Fig. 3. Surface Weather map at 9 AM on June 1, 2022.

Based on Figure 3, the reading from Step 1 included:

“To the south of Japan (below the center of the image): An isobar of 1010hPa on the outer edge of a high pressure area is observed. To the west of Japan (to the left of the center of the image): A 1004hPa isobar is visible, with a cold front crossing the west of Japan. Over Japan (at the center of the image): No prominent pressure system are evident.”

This resulted in a misinterpretation of the 1010hPa high pressure as being to the south of Japan and an overemphasis on the frontal impact to the west. Consequently, an interpretation was made that Western Japan was experiencing unsettled weather. In contrast, during Step 5’s checking process, the following was noted:

“In Chugoku and Shikoku, there are areas with short sunshine durations, but no significant impacts are observed.”

This interpretation, based on the absence of precipitation data for Chugoku and Shikoku, was ambiguous and did not request corrections; hence it was not reflected in subsequent steps, rendering the checking step ineffective. The generated commentary is as follows:

“A low pressure system over the Sea of Japan, cold front passing through Western Japan. Unstable weather with rain in Kushiro, Hokkaido. Sunny in Morioka, Tohoku. Calm and clear skies in Kanto. Intermittent rain and cloudy in Chugoku and Shikoku. Changeable weather in Kyushu. Strong winds and occasional rain in Okinawa. Warm temperatures across the regions, but be cautious of the front.”

In this generated commentary, Chugoku and Shikoku are described as having intermittent rain and cloudiness, indicating that the content from the checking step was not incorporated.

3.3.3. Omission of critical information

In Experiment 4, there were instances where important information was overlooked amidst the execution of many steps. Here we analyze such an instance from January 6, 2022. The surface weather map image for this day is as shown in Figure 4.

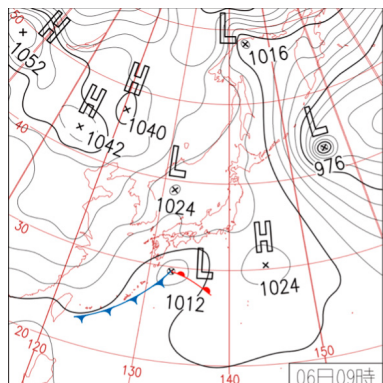


Fig. 4. Surface Weather map at 9 AM on January 6, 2022

January 6, 2022, was a day of snowfall in the Kanto region. The output from Step 1, based on Figure 4, recognized the presence of a low-pressure system along the southern coast. Regarding this, the reading of meteorological data in Step 6 notes:

"In the Kanto region: The data from Mito indicates a precipitation amount of 3.5mm and the deepest snow accumulation, suggesting that the unstable weather continues."

Although there is a mention of snowfall in the Kanto region, it is not strongly emphasized, resulting in the final output stating:

"Due to a strong high pressure directly over Hokkaido, a winter pressure pattern has formed, bringing snow to Hokkaido and the Tohoku region. The southern part of Mainland Japan expects long hours of sunshine and stable weather. Okinawa is prone to rain due to the influence of low pressure."

Thus, the output for the southern part of Mainland Japan, a broad regional classification not commonly used in weather commentary, predicts "long hours of sunshine and stable weather." Consequently, the snowfall in the relatively narrow area of the Kanto region, which has a significant societal impact, was not mentioned.

From these results, it can be considered that, although correct recognition is performed to a certain extent at each step, errors arise because the importance of the information is not interpreted. Therefore, it is thought that improvements could be made by providing prior information on the importance of the information being handled.

4. Discussion

In the recognition of weather map images, ChatGPT displayed a certain level of performance. However, there were also cases where lines on surface weather maps made it difficult to correctly grasp geographical relationships, such as not being able to recognize the difference between a coastline and an isobar. In the university common test, it is assumed that the challenge was due to the black and white images, but there were also cases in weather commentary generation where the model failed to recognize correctly despite coastlines and isobars being in different colors. From these findings, it is thought necessary to explore methods that effectively utilize the characteristics of weather maps, where coastlines, often represented as background images, carry significant information.

In the weather commentary generation experiment, the improvement in scores from Experiment 1 to Experiment 4 indicates the importance of prompt engineering to further optimize AI performance. However, in Experiment 4, a check task was incorporated into the prompts, but it did not work effectively in several cases. This is because in this experiment, the weather map image, weather data, and prompt were sent only once, so the input image was only interpreted in the first step and image recognition was not performed again. It is considered necessary to develop a method to maximize the effectiveness of each step.

5. Conclusion and Future Works

In this research, we verified the recognition performance of surface weather map images using the multimodal AI, ChatGPT4 Vision. It was demonstrated that the system possesses a certain level of recognition capability for answering university common test questions and generating weather commentary. Furthermore, it was shown that presenting meteorological data can correct misrecognitions and contribute to a certain degree of performance improvement where recognition capabilities are lacking. Additionally, it was found that ChatGPT can generate text of a quality comparable to weather commentary created by human meteorologists when given example sentences for natural language generation.

To address the issues raised in the discussion, future research will consider a method of splitting up the prompts. This would improve the effectiveness of the confirmation step and facilitate evaluation of the results of each prompt. Another problem was that the generated comments did not reflect important information such as the amount of snow-fall in Kanto. This could be improved by extracting only noteworthy information from weather conditions in advance, rather than inputting all weather data.

This research demonstrated the potential of multimodal AI in weather map recognition and its practical implications for weather services, education. For example, automated grading systems based on weather map recognition can reduce the workload of educators and provide immediate feedback to students. In addition, AI-generated weather commentary can help weather forecasters create more engaging and informative content for the general public. Advances in multimodal AI, such as those demonstrated in this research, have the potential to revolutionize the way we learn about weather information and generate weather commentary.

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